# Housing First or Treatment First?

A stylized model

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## 1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and roughly forty parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the work. This note strips the environment to the smallest one in which my three main questions are still well posed.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs are a normal good bought at a price, so that concentration is not what a model produces by default. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring the disorder, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, never. Housing First always raises welfare when agents are forward looking, as they are here. Addiction creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension alone cannot justify conditionality. However, if addiction and homelessness generate an externality, such as reducing the earning potential of children, then Treatment First may be preferred. The model's work here is to show why the first claim holds and then to qualitatively price the externality that overturns it.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains is what the results need. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

## 2 The environment

### 2.1 Households and preferences

There are three periods and a continuum of households. A period stands for twenty years of adult life, so the three of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is 0.99 annually and $\beta = 0.99^{20} = 0.818$ per period, the gross interest rate is $R = 1/\beta = 1.223$ so that $\beta R = 1$, and $\phi_0$ is survival over twenty years. Notation follows the quantitative model throughout.

A household is characterised by its permanent productivity $x$, fixed over its life and lognormally distributed across households, $\log x \sim \mathcal{N}(-\frac{1}{2}\sigma^2, \sigma^2)$, so mean income is one. Households differ in $x$, in age $j \in \{1, 2, 3\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in disorder status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption $c$, housing services $h$ and drug consumption $v$,

$$u(c,h,v) \;=\; (1-\gamma-\theta)\log c \;+\; \gamma\log h \;+\; \theta\big[\log(b+v)-\log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so $\gamma$ is the housing budget share proper. Drugs enter only in the first period, since that is the only period in which $v$ is chosen, and housing is chosen each period from a three point grid $\{h_0, h_S, h_L\}$. The pair $(\theta, b)$ governs the drug term: the marginal utility of the $v$th unit is $\theta/(b+v)$, so $\theta$ mostly sets how much a user uses and $b$ whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, since under plain $\theta\log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

## 2.2 Timing

**Period 1.** Every household is employed and clean. It chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent $p_r$, which is normalised to one, on whatever unit it picks. Using drugs generates the risk of a substance use disorder next period, increasing in $v$,

$$\pi(v) \;=\; 1-e^{-\lambda v}. \tag{2}$$

Because $\pi$ is increasing in $v$, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

**Period 2.** For households that used, a disorder $d \in \{0,1\}$ is realised with probability $\pi(v)$; for the rest no disorder is realised. A household without the disorder is unemployed with probability $u_0$, and a household with the disorder is unemployed with the higher probability $u_1$. If unemployed, it receives income support. The household then chooses consumption, housing and saving. The probability of recovering from the disorder by the next period is $\rho$.

**Period 3.** Terminal, after which the household dies with certainty. Employment, income support, the voucher and the housing choice work as in period 2, there is no drug use decision, and assets are run down with nothing bequeathed.

**Mortality, and why there are three periods.** A household survives from one period to the next with probability

$$\phi(d) \;=\; \phi_0\,(1-\kappa d), \tag{3}$$

so $\kappa$ is the proportional reduction in survival caused by the disorder. The third period is what makes that reduction bite. Mortality is a tax on a continuation value, so it can only matter if there is a period after the one in which the disorder is carried. In a two period version the disorder is realised in the last period, death costs nothing because nothing follows it, and the mechanism the whole model rests on is silent. Three periods is the smallest number at which acquiring the disorder has a future to destroy.

## 2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the disorder. Income in a period, given permanent productivity $x$ and employment status $e$, is

$$y(x,e) \;=\; \begin{cases} x & \text{if } e=1, \\ \psi\cdot\min\{x,\bar{y}\} & \text{if } e=0, \end{cases} \tag{4}$$

with $\psi$ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of $\psi$ is what links the disorder to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h \in \{h_0,\ h_S,\ h_L\}, \qquad h_0 < h_S < h_L, \tag{5}$$

homelessness, a small unit and a standard unit. Homelessness is free and delivers almost nothing. Two rented sizes is the fewest that lets a household economise on housing rather than lose it: with one, homelessness comes out at 3.48 per cent against a target of 0.40, at any $h_0$.

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0, 1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction $\kappa_r$ of market rent, so the rent paid on option $j$ is

$$r_j(\eta) \;=\; p_r\, h_j \big[\kappa_r \eta + (1 - \eta)\big] \cdot \mathbf{1}\{j \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in periods 2 and 3, and $\pi_V$, the share of eligible households actually covered, is the policy dial of Section 5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of $(d', v)$. **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since cover in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

### 2.4 The household's problem

Let $a$ be assets carried in. In period 3 the household consumes everything and chooses housing,

$$W_3(a, x, e, \eta) = \max_j \Big\{(1 - \gamma - \theta)\log c + \gamma \log h_j + \aleph\Big\}, \qquad c = Ra + y(x, e) - r_j(\eta). \tag{8}$$

In period 2 it also saves, and faces mortality and recovery,

$$W_2(a, x, e, \eta, d) = \max_{j,\, a' \ge 0} \Big\{(1 - \gamma - \theta)\log c + \gamma \log h_j + \aleph + \beta\, \phi(d)\, \mathbb{E}\, W_3(a', x, e', \eta')\Big\}, \tag{9}$$

with $c = Ra + y(x, e) - r_j(\eta) - a'$, and the expectation taken over next period's employment and voucher draws, which depend on the disorder status $d'$, itself equal to zero with probability $\rho$ when $d = 1$ and to $d$ otherwise. In period 1 the household is employed, clean and holds no voucher, so its state is $x$ alone, and it chooses housing, drug use and saving,

$$V_1(x) = \max_{j,\, v \ge 0,\, a' \ge 0} \Big\{(1 - \gamma - \theta)\log c + \gamma \log h_j + \theta[\log(b + v) - \log b] + \aleph + \beta\, \phi_0\, \mathbb{E}\, W_2(a', x, e', \eta', d')\Big\}, \tag{10}$$

with $c = x - r_j(0) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). A household uses if the value at its best positive $v$ exceeds the value at $v = 0$, and because households differ only in $x$ that defines a set of income intervals on which use occurs.

## 3 Calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the annual discount factor is 0.99 and the gross interest rate is $1/\beta$, the dispersion of log permanent income is $\sigma = 0.75$ from Heathcote, Perri and Violante (2010), and per period survival is $\phi_0 = 0.93$ from the CDC life tables. For the disorder, recovery over a period is $\rho = 0.40$ from the SAMHSA treatment episode data, the proportional fall in survival is $\kappa = 0.45$ from Hjorthøj et al. (2023), and the two unemployment probabilities are measured directly, $u_0 = 0.049$ from the BLS and $u_1 = 0.1625$ from NSDUH 2024 for respondents aged 26

Table 1: Externally set parameters

| Parameter | Description | Source or rationale | Value |
|---|---|---|---|
| *Preferences and demographics* | | | |
| $\beta$ | Discount factor over a twenty year period | 0.99 annually, compounded | 0.818 |
| $R$ | Gross interest rate over a twenty year period | Set to $1/\beta$ | 1.223 |
| $\sigma$ | Dispersion of log permanent income | Heathcote, Perri and Violante (2010) | 0.75 |
| $\phi_0$ | Survival per period, no disorder | CDC life tables | 0.93 |
| $\theta$ | Weight on drug use | Set; see the text | 0.11 |
| $\gamma$ | Weight on housing services | $0.25(1-\theta)$, Consumer Expenditure Survey | 0.223 |
| *The disorder* | | | |
| $\rho$ | Recovery probability per period | SAMHSA, TEDS | 0.40 |
| $\kappa$ | Proportional fall in survival with the disorder | Hjorthøj et al. (2023) | 0.45 |
| $u_0$ | Unemployment probability, no disorder | BLS | 0.049 |
| $u_1$ | Unemployment probability, with the disorder | NSDUH 2024, ages 26 and over, weighted | 0.1625 |
| *Income support* | | | |
| $\psi$ | Replacement rate while unemployed | DOL, statutory UI rates | 0.50 |
| $\bar{y}$ | Cap on the earnings base, multiple of average income | DOL, state benefit maxima | 1.00 |
| *Housing, drugs and the voucher* | | | |
| $p_r$ | Rent per unit of housing services | Normalisation | 1.00 |
| $p_v$ | Price of a unit of drugs | Normalisation of the units of $v$ | 1.00 |
| $h_L$ | Standard rental unit | ACS 2023, rent over income | 0.212 |
| $h_S$ | Small rental unit | Half the standard unit | 0.106 |
| $\kappa_r$ | Tenant's share of the rent when covered | HUD voucher rules | 0.30 |
| $y_V$ | Voucher means test, multiple of average income | HUD income limits | 1.00 |
| $\pi_V$ | Voucher generosity in the baseline | HUD, share of eligible assisted | 0.27 |

and over. Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The standard rental unit costs 21.2 per cent of mean income, its rent to income ratio in the ACS, and the small unit is half of it. The voucher parameters are institutional: covered tenants pay 30 per cent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the HUD share of eligible households actually assisted.

Four parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly: with no preference heterogeneity the aggregate is a cliff in $b$, so a gradient method steps out of the region where participation is interior and walks to a corner. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit $b$ is matched to a drug participation rate of 5.13 per cent, housing services on the street $h_0$ to a homelessness rate of 0.399 per cent from the HUD point in time counts, and the addiction hazard $\lambda$ to the 33.2 per cent of users who acquire a disorder.

The weight on drugs $\theta$ is set rather than calibrated, and the reason is worth stating because the obvious alternative was tried. The natural target is the 11 per cent of income that users who are not addicted spend on drugs, which the quantitative model matches, and this model cannot reach it while keeping users poor. Two moments already pin the pair $(\lambda, v)$: the disorder rate among users, $1 - e^{-\lambda v} = 0.332$, fixes $\lambda v$ at 0.403, and an 11 per cent spending share on an average user income near 0.2 would fix $v$ near 0.022 and so $\lambda$ near 18. Scanning $(\lambda, \theta)$ over that region shows the model is bimodal with no interior solution: below a threshold in $\theta$ users are the very poor and spend a hair above nothing, above it users are the rich and the income gradient the model exists to explain is gone. The economics is not subtle. What puts use at the bottom is that the disorder destroys a continuation value, and for a poor household that cost is enormous relative to the utility of the drug, so it uses as little as it can while still using; low intensity use among the poor is the same fact as the income gradient here, not an independent one. Setting $\theta = 0.11$ and $\gamma = 0.25(1-\theta) = 0.223$, the model produces a spending share of 1.76 per cent. It misses that moment by a wide margin, and the quantitative model, with its separate light and heavy experimentation margins and its drug demand from households that already carry the disorder, is where it should be matched.

Table 2: Internally calibrated parameters

| Parameter | Description | Target moment | Data | Model | Value |
|---|---|---|---|---|---|
| $\aleph$ | Flow value of being alive | Value of a statistical life, annual incomes | 80.0 | 80.0 | 2.855 |
| $b$ | Pull of the first unit of drugs | Drug participation rate | 5.13% | 5.13% | 0.00032 |
| $h_0$ | Housing services when homeless | Homelessness rate | 0.399% | 0.371% | 0.00073 |
| $\lambda$ | Addiction hazard per unit of use | Share of users who acquire the disorder | 33.2% | 36.3% | 783.1 |

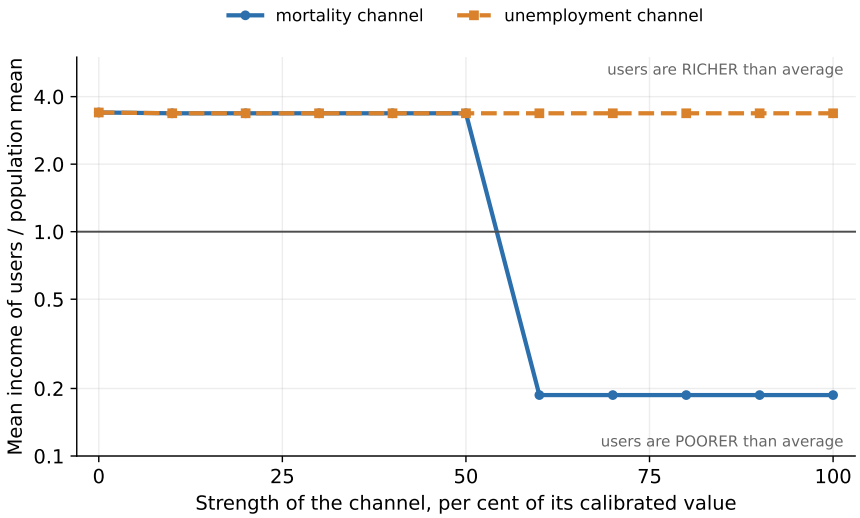



Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and $b$ is recalibrated at every point so that participation is 5.13 per cent throughout.

# 4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a disorder that is the only force operating, and drug use concentrates at the top: the average user then earns 3.40 times the population average.

Two features of the environment could reverse that. The *mortality channel* is that the disorder kills you, through $\kappa$ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to its calibrated value, recalibrating $b$ at every point so participation stays at 5.13 per cent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

The result is sharper than the statement that mortality matters more. That channel reverses the gradient at half of its calibrated strength, taking the income ratio from 3.40 to 0.19, while the unemployment channel does not reverse it at all, leaving users at 3.37 times average income even fully switched on. A resource penalty cannot produce the concentration of drug use at the bottom.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

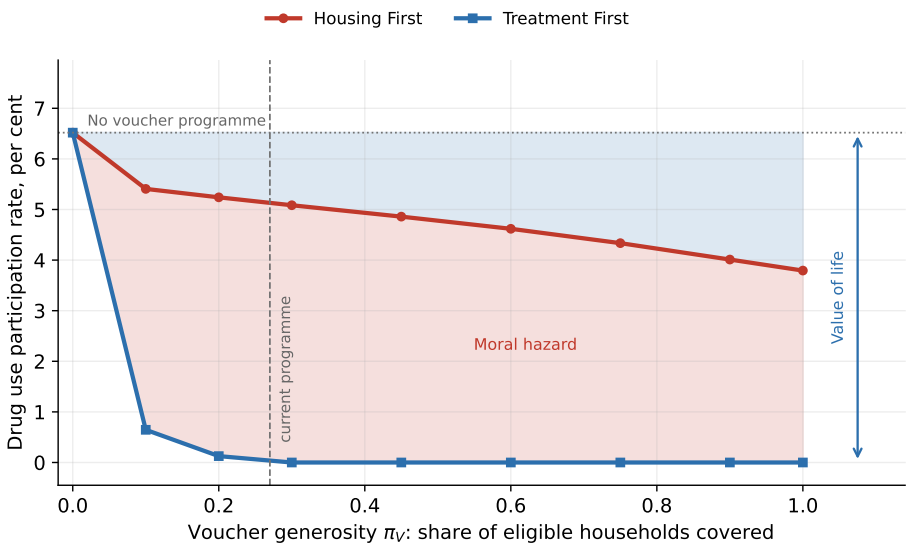



Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

# 5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

Figure 2 traces both as generosity rises. With no programme at all, 6.52 per cent of households use. Universal Treatment First takes that to 0.00 per cent, a value of life force of -6.52 percentage points; universal Housing First leaves it at 3.79 per cent, so the moral hazard force is +3.79 points and the net effect of Housing First is -2.73. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and disorder outcomes for each programme as its generosity changes, alongside drug use. On homelessness the two policies end in the same place, both driving it from 0.78 per cent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the disorder that puts their housing at risk. The larger difference is on the other margin, where at universal cover Housing First leaves 0.602 per cent of the population carrying the disorder and Treatment First leaves 0.000 per cent. What the moral hazard buys is extra disorder cases among households that are housed either way.

# 6 Welfare

## 6.1 Without externalities, Housing First always wins

Sections 4 and 5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity $\pi_V$, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a

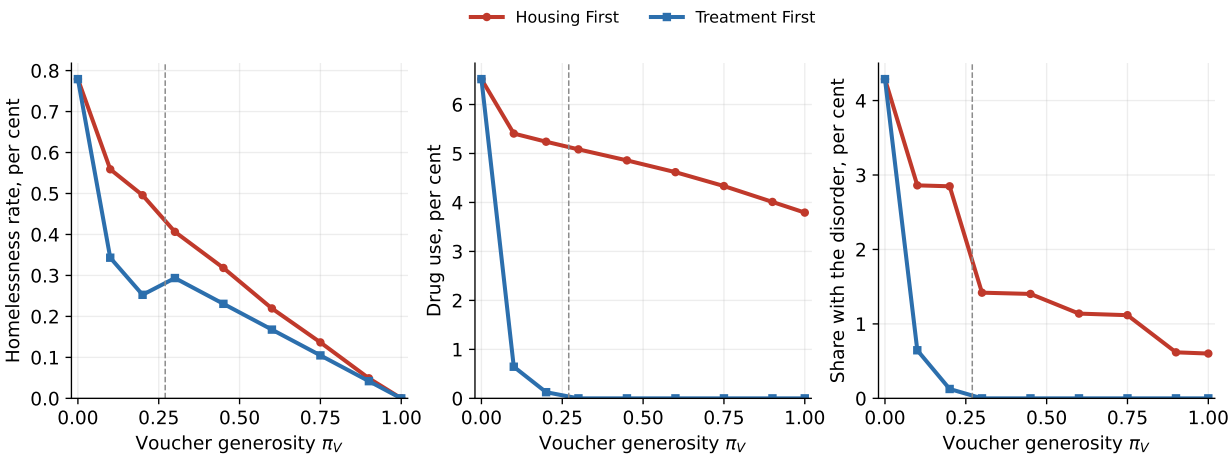



Figure 3: Homelessness, drug use and the disorder under the two policies

Table 3: Welfare at universal cover, with no externality

| | Housing First | Treatment First |
|---|---|---|
| *Panel A. Outcomes at universal cover* | | |
| Drug participation rate | 3.79% | 0.00% |
| Share with the disorder | 0.602% | 0.000% |
| Homelessness rate | 0.000% | 0.000% |
| Voucher outlay per household | 0.1694 | 0.1696 |
| *Panel B. Treatment First against Housing First, no externality* | | |
| Households strictly worse off | | 4.1% |
| Households strictly better off | | 0.0% |
| Consumption equivalent loss, those affected | | 0.41% |
| Consumption equivalent loss, whole population | | 0.017% |

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household from period 2 on, discounted, per unit of average income.

household that optimises cannot be worse off:

$$V_1(x;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(x;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } x, \tag{11}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[1]

It follows that the moral hazard is not a distortion. Households here face the true addiction hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +3.79 points of Section 5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 4.1 per cent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.41 per cent of consumption.

A fixed budget does not rescue conditionality either. Treatment First is actually slightly *more* expensive at universal cover, 0.1696 against 0.1694, so matching on cost rather than on generosity changes nothing. The reason is that survival to period 3 falls from 0.93 to 0.51 with the disorder, so the households the moral hazard adds to the rolls die sooner and draw voucher payments for fewer periods. So a case for conditionality has to come from an externality.

[1] The same comparison in the quantitative model gives the same answer.

Table 4: The externality on children

| *Panel A. Transmission: what each income penalty implies* | | | |
|---|---|---|---|
| $\delta_d$ | Child's disorder rate | Ratio | Loss to the child |
| 0.00 | 2.20% | 1.00 | 0.0% |
| 0.20 | 3.95% | 1.80 | 21.7% |
| 0.30 (calibrated) | 5.34% | 2.43 | 37.3% |
| 0.40 | 7.27% | 3.31 | 58.2% |
| *Panel B. Does the externality reverse the ranking?* | | | |
| Private welfare loss from conditionality | | | 0.017% |
| Disorder cases prevented | | | 0.602 pp |
| Damage to the child of one such case | | | 34% |
| Externality gain from conditionality | | | 0.202% |
| Gain over loss | | | 12.1 |

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Losses are consumption equivalents.

## 6.2 The externality on children reverses it

The externality I model is one that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste, a lower $b$, matches the transmission fact but has the wrong welfare sign: $b$ enters only through $\theta \log(1+v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. Letting it inherit a greater susceptibility, a higher $\lambda$, fails the other way, since a child facing a riskier drug optimally uses *less* and is worse off ex ante, which is clearly counterintuitive.

Income works on both counts, and it works through the channels explored in Section 4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. The income gradient does the whole job, so the model transmits addiction across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$x_{\text{child}} \;=\; x_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the disorder}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{12}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into $V_1$.

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the disorder 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. The homelessness penalty $\delta_h$ is set directly from Chetty, Hendren and Katz (2016), who find that children who moved to a low poverty neighbourhood before age 13 went on to earn 31 per cent more as adults. The two penalties also do different work: both policies drive homelessness to zero at universal cover, so $\delta_h$ prices the programme against no programme while $\delta_d$ decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.017 per cent of consumption, small because only 4.1 per cent of households are affected. Against that, conditionality prevents 0.602 percentage points of disorder cases in the second row, and each of those cases costs one child 33.6 per cent of consumption, in the third. The externality gain is 12.1 times the private loss, and stays above it throughout the range of $\delta_d$ in Panel A. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

Treatment First drives drug use to *exactly* zero here, and that inflates the margin. Nobody in this model faces the sobriety condition and fails to meet it, and the households who cannot comply are the entire empirical case for Housing First. Establishing which way each comparison goes, with those households, a government budget, an equilibrium drug price and a proper life cycle, is what the quantitative model is for.

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